

Chef MAI: The MiniChef AI

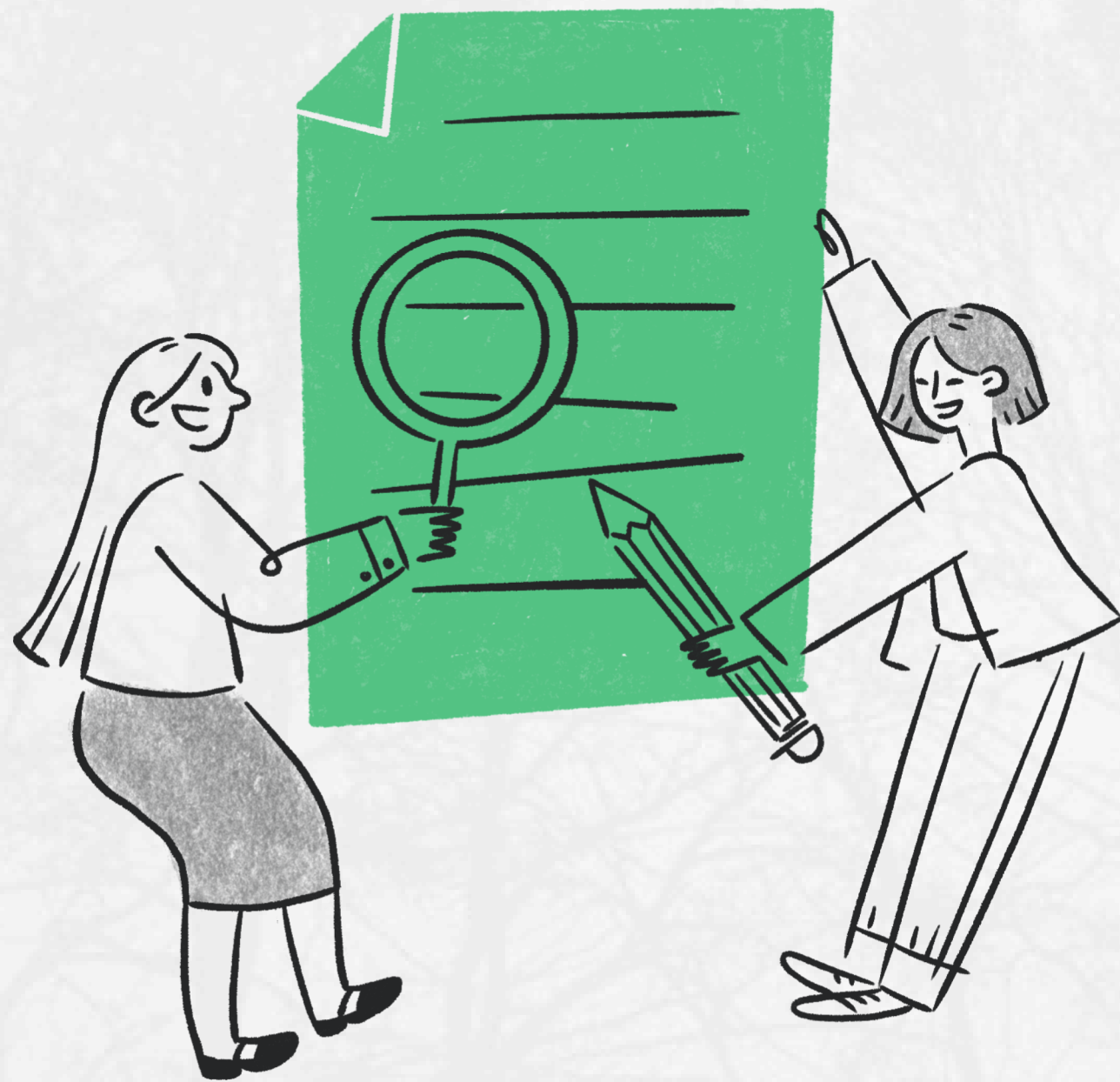
**An Ingredient-Based Smart Recipe Recommendation Application
Using Bayesian Learning and Relevance-Based Matching**

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Abstract



- Cooking with limited ingredients and a lack of prior knowledge and guidance often leads to household food waste.
- **Chef MAI: The MiniChef AI** suggests recipes based on available ingredients.
- The system uses **Bayesian Learning and Relevance-Based Matching** to filter and recommend suitable recipes.
- Users can catalog ingredients through text or image to receive tailored meal options and smart substitutions for missing items.

Question

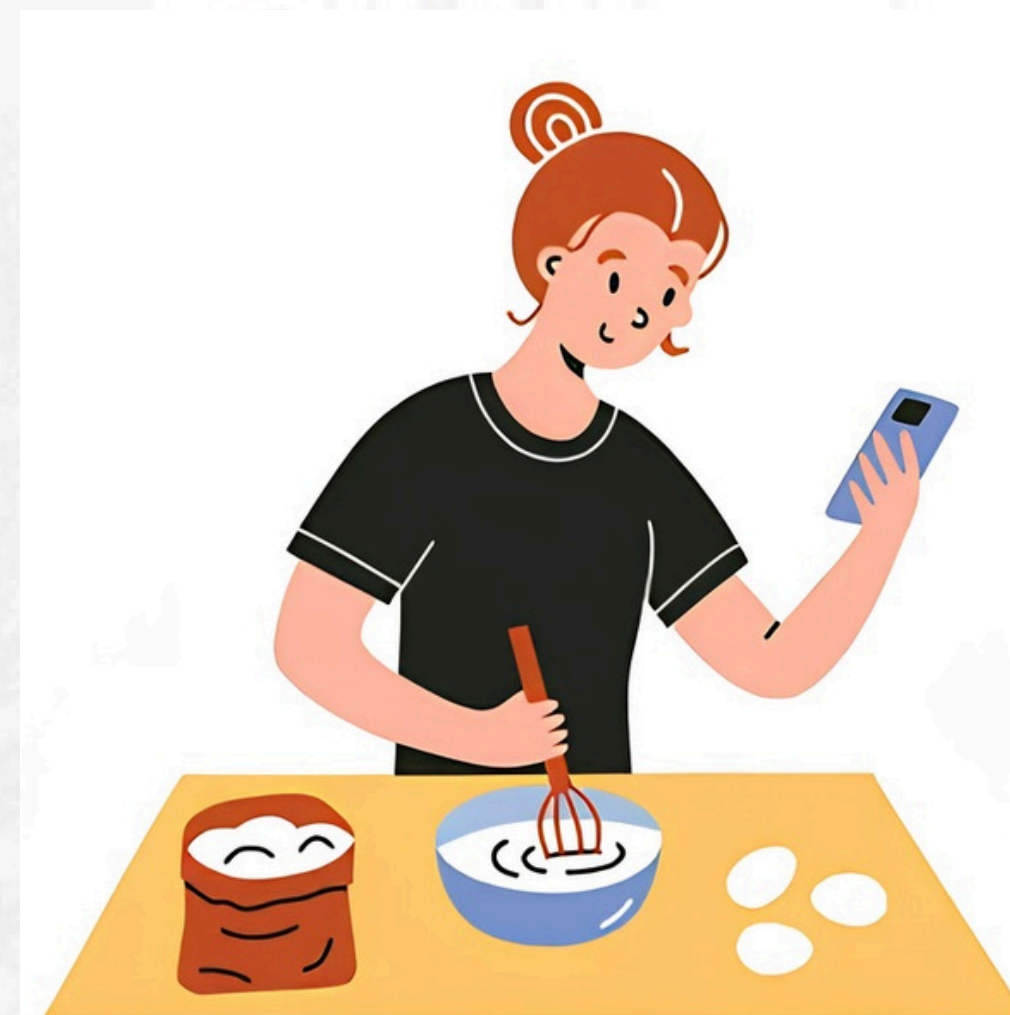
- How can Learning from Examples, Bayesian Learning, Naive Bayes Classification, and Learning Using Relevance Information be applied to recommend recipes that match a user's available ingredients, preferences, and cooking constraints?

Hypothesis

- Using Bayesian Learning, Naive Bayes Classification, and Relevance-Based Recipe Matching will allow Chef MAI to recommend recipes that better match the user's available ingredients, cooking preferences, and behavior compared to simple keyword-based recipe searching.

Background

- Chef MAI is a mobile recipe recommendation app that **helps users decide what to cook using ingredients they already have.**
- It addresses common problems such as:
 - limited cooking knowledge
 - leftover ingredients, food waste
 - time-consuming recipe searches.
- The app allows users to input ingredients, filter recipes by cuisine, time, difficulty, and preferences, and receive personalized meal suggestions.



PEAS

PERFORMANCE MEASURE

- Relevant recipes
- High ingredient match
- High user satisfaction

ENVIRONMENT

User's kitchen or pantry

ACTUATORS

Screen Display for Recipe Recommendation

SENSORS

- Keyboard (Text ingredient input)
- Camera (Image input)

Procedure

1. The user inputs ingredients manually or through the "Take a Pic!" feature.
2. The system processes recipe data such as ingredients, cuisine, cooking time, difficulty, and ratings.
3. Relevance-based matching prioritizes the most important factors, especially available ingredients and user preferences.
4. Naive Bayes Classification estimates which recipes are most suitable.
5. Bayesian Learning updates the user's taste profile based on likes, skips, ratings, and favorites.
6. The app displays recommended recipes with ingredients, substitutes, cooking time, difficulty, and directions.

Metrics

- **Accuracy / Suitability Rate**

- Measures how often the system recommends recipes that are actually suitable for the user.

$$\text{Suitability Rate} = \frac{\text{Number of Suitable Recommendations}}{\text{Total Recommendation}}$$

Metrics

- **Precision@K**

- Measures how many of the top recommended recipes are relevant.

$$\text{Precision@K} = \frac{\text{Relevant Recipes in Top K}}{K}$$

Metrics

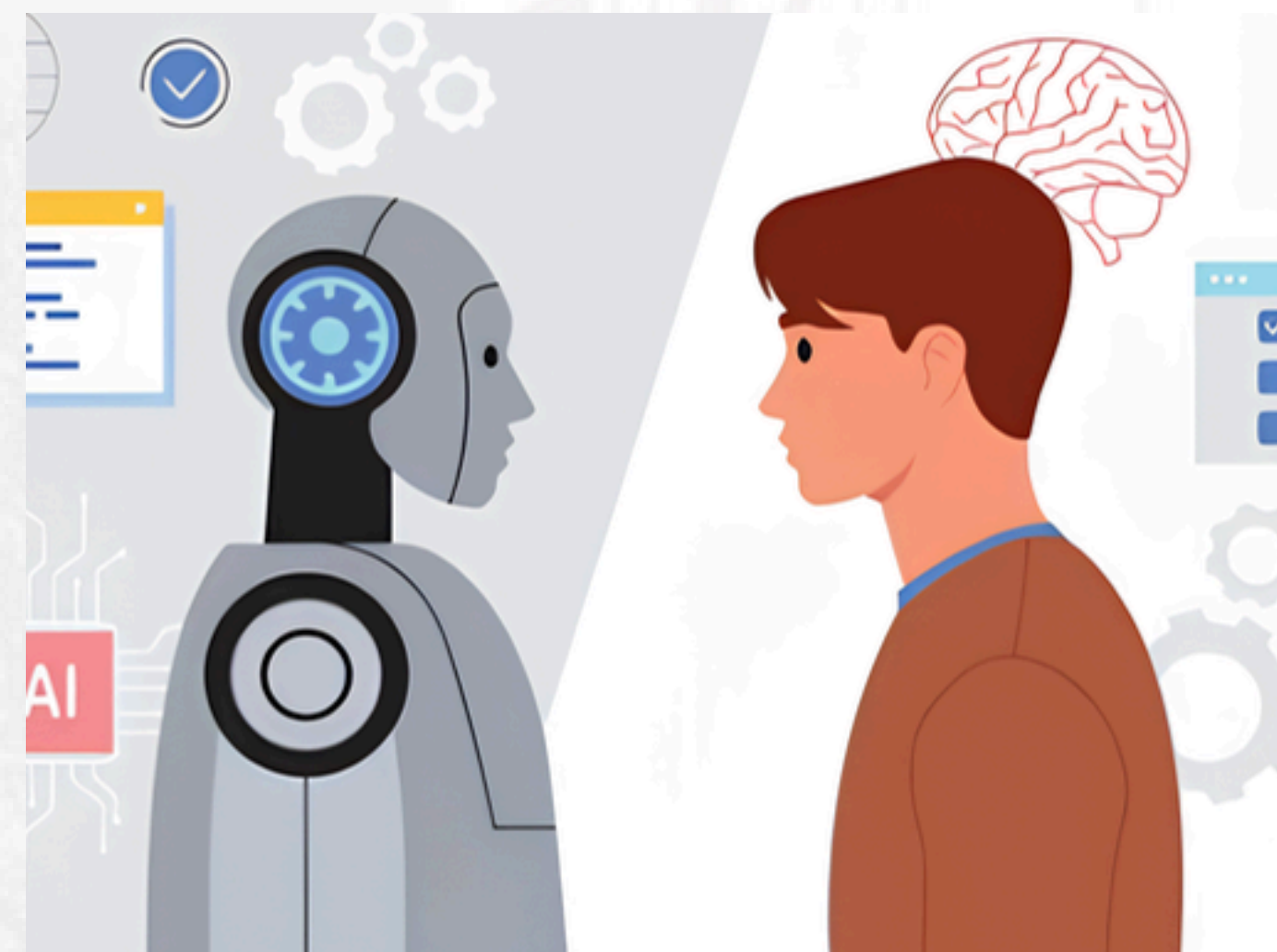
- **Ingredient Match Score**

- Measures how well a recommended recipe matches the ingredients that the user already has.

$$\text{Ingredient Match Score} = \frac{\text{Available Ingredients Used in Recipe}}{\text{Total Ingredients Required by Recipe}}$$

Discussion

- **Chef MAI** works as an **intelligent recipe assistant**. Instead of only searching for recipes by keyword, it learns from recipe examples and user interactions.
- Learning from Examples **learns from recipe records and user behavior**.
- Bayesian Learning **updates user preferences over time**.
- Naive Bayes Classification **classifies recipes as suitable or unsuitable**.
- Relevance-Based Recipe Matching **prioritizes important recipe features**.



Bibliography

- Guo, S. Bayesian Recommender Systems: Models and Algorithms, 2011. <https://users.cecs.anu.edu.au/~sguo/thesis.pdf>
- Rodrigues, M. S., et al. RecipeIS: Recipe Recommendation System Based on Recognition of Food Ingredients, 2023. <https://www.mdpi.com/2076-3417/13/13/7880>
- Chow, Y. Y. Food Recommender System: A Review on Techniques, Applications and Evaluation Metrics, 2023. <https://www.aasmr.org/jsms/Vol13/No.5/Vol.13%20No.5.10.pdf>
- Garrido-Merchán, E. C., & Albarca-Molina, A. (2018). Suggesting cooking recipes through simulation and Bayesian optimization. Lecture Notes in Computer Science, 277-284. https://doi.org/10.1007/978-3-030-03493-1_30
- Gim, M., Park, D., Spranger, M., Maruyama, K., & Kang, J. (2021). RecipeBowl: A cooking recommender for ingredients and recipes using set transformer. IEEE Access, 9, 143623-143633. <https://doi.org/10.1109/access.2021.3120265>



THANK YOU FOR LISTENING!

Do you have any questions?